# Pollen Viability Staining

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<u>Preparation</u>

Alcohol

Chloroform

Acetic Acid, Glacial Acetic Acid

Glycerol

Malachite green (1% solution in 95% alcohol)

Acid fuchsin (1% solution in water)

Orange G (1% solution in water)

D.W.

<u>Fixative (50ml)</u>

Carnoy's fixative = alcohol : chloroform : acetic acid = 6 : 3 : 1 = 30ml : 15ml : 5ml

<u>Stain Solution</u>

25 mL Distilled water

12.5 mL Glycerol

5 mL 95% alcohol

2.5 mL Acid fuchsin (1% solution in water)

2 mL Glacial acetic acid

500λ Malachite green (1% solution in 95% alcohol)

250λ Orange G (1% solution in water)

+ Rest: D.W.

Procedure

1. Fixation

- minimum: 2 hours / maximum: 12 months (either RT or cold room)

- Choose freshly harvested flower buds that are about to open with non-dehiscent anthers.

-NOTE: Do not take younger flower buds because Alexander staining is intended for mature pollen grains and does not efficiently stain immature pollen grains.

2. Place sample on slide glass. fixative's liquid was thoroughly and carefully dried from the plant material with absorbent paper.

3. Apply 2-4 drops of the stain solution before the sample completely dries.

4. Once the sample is in the stain, slowly heat the slide over an alcohol burner in a fume hood until the stain solution is near boiling (~30 seconds).

- A more moderate rate of heating allows better penetration of the dye into the cellulose and protoplasm of the pollen.

- Extremely high temperatures resulting in smoking or bubbling of the stain can burn the dye and the sample.

- Heating can be adjusted by briefly moving the slide in and out of the flame.

5. Washing with D.W. carefully

6. Mount cover glass with D.W.

Cautions

1. Proper safety gloves should be worn to avoid the risk of chloroform from being absorbed through the skin.

2. To ensure stain has been completely absorbed into the pollen grains, 10 to 15 minutes should be allowed for some species such as Lonicera tatarica, Ginkgo biloba, Pinus resinosa and Rhodo dendron mucronulatum

Reference

[1] Peterson, Ross, Janet P. Slovin, and Changbin Chen. "A simplified method for differential staining of aborted and non-aborted pollen grains." *International Journal of Plant Biology* 1.2 (2010): e13.